

Corda: The Smart Contract Based Blockchain Project

Making reliable transactions over a network is the baseline requirement of most businesses. Blockchain technology enables different parties to transact efficiently and in a verifiable manner. Corda is an open source blockchain project. Its smart contract technology allows businesses to transact safely. This article introduces you to its features.



With the massive increase in the size and nature of services that are provided through the networks, security has emerged as a priority requirement. Even as there are innumerable efforts on to improve transaction safety, the blockchain has captured people's attention in recent times. By design, the blockchain is resistant to data modification. Primarily, the blockchain consists of a growing list of records termed as blocks. These blocks are linked using cryptography. Each block in the chain holds the cryptographic hash of the previous block, a timestamp, in addition to the transaction data. The Merkle tree structure is used to represent this. Satoshi Nakamoto (true identity unknown) is the person or the name used by a group of people who conceptualised the blockchain in the year 2008.

The blockchain is used across a spectrum of domains these days, and there are many projects/platforms based on it (Figure 1).

This article explores an important open source blockchain project called Corda, which has been developed

by R3 along with the participation of 200 technology and industry partners.

The primary features of Corda (Figure 2) are listed below:

- Java/JVM language based smart contracts.
- The availability of the Flow framework, which enables managing communication and negotiation between parties.
- Provision of a notary infrastructure to validate transactions without global broadcast.
- Enables the development of CorDapps.
- Built using Kotlin targeting JVM.

A simple introduction to Corda is available in the two-minute video provided by the official Corda documentation <https://vimeo.com/205410473>.

The terminology associated with Corda is vast and covering the entire list of terms would be impossible in this article due to space constraints. Here are some of the most commonly used terms:

- The network
- The ledger